



Metrics on MDPs

Which luckily enable policy transfer

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 - ❓ Goal
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Background

An MDP is a tuple $(X, A, \{P_{ss'}^a\}, \{r_s^a\})$

- ❑ X is the state space;
- ❑ A is the action space;
- ❑ P is a probability;
- ❑ r is a reward.

A policy π maps states to actions.

V^π is the value of the policy.

There exists an optimal policy π^* with value V^* and we can compute it.

Metrics for finite MDPs (2004)

Problem: Some environments are too complex

Idea: Simplify the environment to have smaller sub-problems

Goal: Figure out a definition of similarity between states in an MDP

Solution: A metric of similarity between states

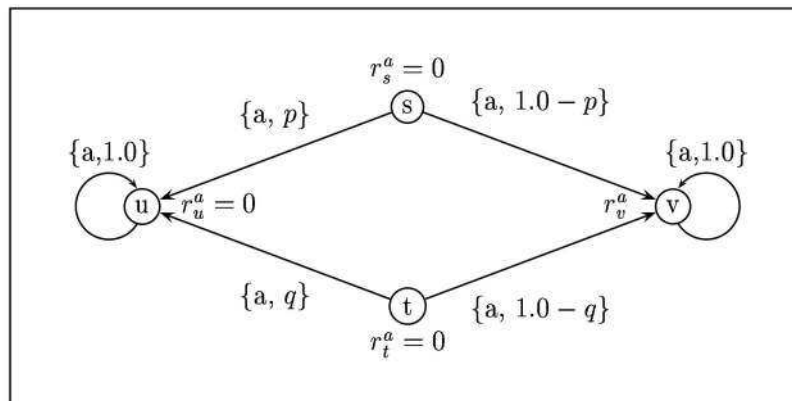
Most natural choice: an equivalence relation between states

Definition: Bissimulation relation

- Two states are *bissimilar* iff
 - For all actions a
 - The rewards are equal
 - The probability of going to any set of equivalent states is equal
- We use the notation $s \sim s'$.

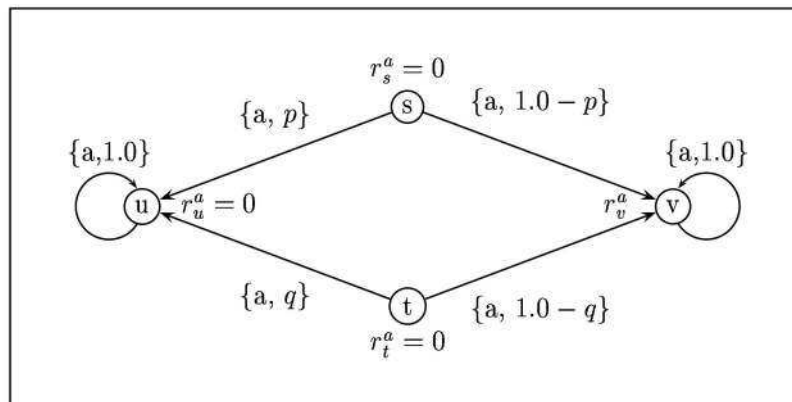
Goal: Figure out a **definition** of similarity between states in an MDP

Example



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Example



Drawback: it is too hard to be bisimilar!

The distance between any pair of states should consider (for all actions):

- The differences in rewards
- The differences in transition probabilities

Definition: Bissimulation metric

- Metric d that induces the bissimulation relation
 - That means $s \sim s'$ iff $d(s, s') = 0$

Theorem:

- There exists a bisimulation metric
- We can compute it in polynomial time
- It is such that:

$$d(s, s') = \max_a \left(\left| r_s^a - r_{s'}^a \right| + \gamma T_K(d) \left(P_s^a - P_{s'}^a \right) \right)$$

Corollary:

- For each ϵ , we cluster the states at distance $< \epsilon$
- We then have a new (reduced MDP) where each cluster C is a state
- For any cluster C and any state s in it:

$$\left| V^{\cdot}(C) - V^{\cdot}(s) \right| \leq \frac{2\epsilon}{c_R(1-\gamma)}$$



Conclusion:

- We have a way of measuring similarity between states
- We can use it to reduce MDPs and solve a smaller instance

Approximate MDP homomorphisms (2008)

Problem: Action similarities were not taken into account

Idea: Take action similarities into account

Goal: Define another metric

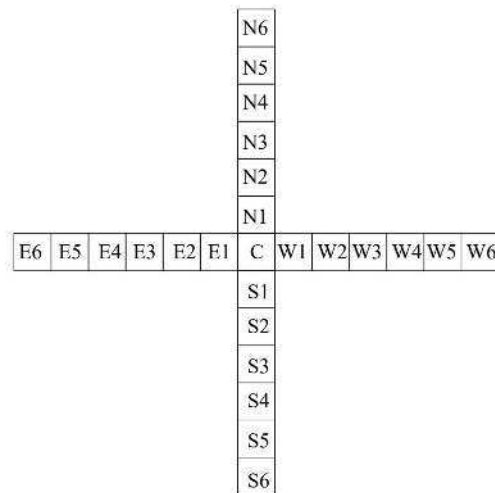
Solution: A metric

Definition: Lax-bissimulation relation

- Two states are *lax-bissimilar* if
 - For all actions a
 - There exists an action b such that
 - The rewards are equal
 - The probability of going to any set of equivalent states is equal
- We use the notation $s \sim s'$ again.

Goal: Figure out a **definition** of similarity between states in an MDP

Example





Theorem:

- There exists a lax-bissimulation metric d
- We can compute it in polynomial time

Corollary:

- In particular, we can solve the MDP of clusters obtained from d:
 - For each $\mathcal{E}$, we cluster the states at distance $< \mathcal{E}$
 - We then have a new (reduced MDP) where each cluster C is a state
 - For any cluster C and any state s in it:

$$\left| V \cdot (C) - V \cdot (s) \right| \leq \frac{4\mathcal{E}}{c_R (1 - \gamma)}$$



Conclusion:

- We can solve the aggregated MDP, which is simpler

Bissimulation for policy transfer (2010)

Question: Can we use the above for different MDPs?

Answer: Yes



We know there is lax-bissimulation relation between MDPs, $\sim$

- We also know there is a lax-bissimulation metric d

Suppose we know the lax-bissimulation metric (we can compute it...)

- We know the distances between states in different MDPs
- We know the distances between state-action pairs in different MDPs



Suppose we have computed a policy π in MDP 1 and want to know how to act on MDP 2.

Algorithm:

- For each state t on MDP 2
 - We choose the closest state s on MDP 1
 - We choose the action on MDP 2 minimizing the distance between $(s, \pi(s))$ and $(t, \cdot)$



Theorem:

The transferred policy is good if the states are close:

$$\left| Q^{\cdot}(t, \pi(t)) - V^{\cdot}(t) \right| \leq 2 \min_s d(s, t)$$



That's it

Definition: Homomorphism

- Map $h = (f, \{g_s\})$ between two MDPs where
 - f maps the states
 - g maps the actions
- h is such that
 - Rewards are matched by h
 - Probabilities are matched by h



Theorem:

- There a bissimulation relation iff
 - there is an MDP homomorphism

Definition: aggregated MDP

- Given an MDP and an homomorphism, an aggregated MDP is such that:
 - The state space is a partition of the original
 - Actions are relabelled

Definition: d-consistent MDP

- An aggregated MDP is d-consistent if
 - Within each class
 - We can find a state s such that
 - For all other states s' in the class and all actions
 - The distance in the aggregated MDP with actions relabelled is smaller than the original distance.

Theorem:

- If an aggregated MDP is d-consistent then for all original states (mapped to a state C)
 - We can solve the aggregated MDP instead